Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/28/2021
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Biofilm Formation, Growth, and Gene Expression on Different Materials and Environmental Conditions in Microgravity		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Luis.Zea@Colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	407-242-2885
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Aerospace Engineering Sciences		
PI Address 2:	429 Ucb, ECAE 1B02		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	01/03/2017	**End Date:**	06/02/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
Contact Email:	louise.s@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 6/02/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 1/02/2021 per S. Reagan/MSFC (Ed., 10/15/19)		
Key Personnel Changes/Previous PI:	July 2021 report: Co-I added (for the reflight): Pamela Flores		
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Moeller, Ralf Ph.D. (Deutsches Zentrum Fuer Luft- Und Raumfahrt E.V., Germany) Muecklich, Frank Ph.D. (Saarland University, Germany) Stodieck, Louis Ph.D. (University of Colorado, Boulder) Flores, Pamela B.S. (University of Colorado Boulder)		
Grant/Contract No.:	80NSSC17K0036 ; NNX17AC21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biofilm formation and its concomitant risk of causing equipment malfunction and human illnesses is a problem that needs to be addressed to enable safe long-term human space exploration. With that objective, this proposal aims to characterize biofilm growth during one International Space Station (ISS) increment in terms of mass, thickness, morphology, and associated gene expression using different spaceflight-relevant microbial species and material substrata. This project also aims to elucidate the biomechanical and transcriptomic mechanisms involved in the formation of the "column-and-canopy" biofilm architecture observed in space. The experiment is designed to assess the expression of genes associated with conferring microorganisms with resistance to oxidative stress, acidity, and antimicrobials in biofilms formed in space, and to determine physical mechanisms of material/microorganisms interaction in biofilms. The experiment will be conducted using BioServe's BioCells housing substrata coupons exposed to different microorganisms and media. Post-flight analyses of fixed samples include confocal laser scanning microscopy and RNA-sequencing. The significance of this project stems from gaining valuable new knowledge that may lead to improved methods and/or material properties for controlling biofilm formation during long duration space missions. The selection of bacterial species, growth media, and substrata material make the results applicable to a) the safety and maintenance of crewed spacecraft, b) planetary protection, c) mitigation of biofilm-associated illnesses on the crew, and d) biofilm-associated illnesses on Earth, thus making it relevant not only to this Research Announcement, but also to NASA's Human Exploration and Operations Directorate, Space Biology Program, and Human Research Roadmap. NOTE Sept 2021: Project continues with new grant and same Principal Investigator, Dr. Luis Zea--"Characterization of Fungal Biofilm (Molds) Formation, Growth, and Gene Expression on Different Materials in Microgravity," grant 80NSSC21K1950.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biofilms are associated with multiple illnesses, including endocarditis (bacterial infection of cardiac tissue), cystitis (a urinary tract infection), and otitis media (an inflammatory disease of the middle ear), as well as several other hospital-acquired infections. Understanding how these biofilms form and their relationship with resistance to antibiotics can help us better prepare novel solutions to these problems on Earth.
Task Progress:	1. Both fungal and bacterial experiment preparation procedures were performed and full Space Biofilm experiment was handed over to NASA on October 31st 2019 for NG-12 launch on November 2nd. 2. Bacterial flight and ground operations were successful. Ground samples followed the same timeline in an asynchronous fashion. 3. Bacterial samples returned to Earth and were delivered to science team on January 10th 2020. 4. Post-flight data acquisition for bacterial side has been completed. 5. Preliminary results were presented to NASA. 6. The fungal experiment is scheduled to fly on SpaceX-24 (reflight). 7. Trainings and skills practice for the fungal experiment have already begun in preparation for SpaceX-24. NOTE Sept 2021: Project continues with new grant and same Principal Investigator, Dr. Luis Zea--"Characterization of Fungal Biofilm (Molds) Formation, Growth, and Gene Expression on Different Materials in Microgravity," grant 80NSSC21K1950. See that project for future reporting.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	Flores P, Schauer R, McBride SA, Luo J, Hoehn C, Doraisingam S, Widhalm D, Chadha J, Selman L, Mueller DW, Floyd S, Rupert M, Gorti S, Reagan S, Varanasi KK, Koch C, Meir JU, Muecklich F, Moeller R, Stodieck L, Countryman S, Zea L. "Preparation for and performance of a Pseudomonas aeruginosa biofilm experiment on board the International Space Station." Acta Astronaut. 2022 Oct;199:386-400. https://doi.org/10.1016/j.actaastro.2022.07.015 , Oct-2022
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